

Andre Tran

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EDUCATION

University of Cincinnati

Expected April 2028

Bachelor of Science in Computer Engineering, Minor in Computer Science

GPA: 3.71/4.00

Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Design & Development, Cybersecurity, Digital Design & Digital Systems, Artificial Intelligence, Context Engineering

Affiliated: CodePath, University of Cincinnati Powerlifting Team, University of Cincinnati Bodybuilding Club

EXPERIENCE

Caphne - Study Buddies Matchmaking (caphne.co)

01/2026 - Present

Lead Software Engineer

Ho Chi Minh, Vietnam

- Engineering an end-to-end distributed real-time messaging system using Socket.IO, PostgreSQL, and Redis, cutting average API response times by 60% and sustaining 1.7K+ requests/minute over 30 days of production traffic.
- Scaled a backend study-matching platform to 400+ active users by leading 6 engineers through sprint planning and code reviews using Nuxt 3 and Node.js across the full product lifecycle.
- Achieved p50 median response times under 400ms by architecting a Vue-based frontend with optimistic updates, lazy loading, and input debouncing, hardened with JWT, OAuth 2.0, and Typebox schema validation.

KatanaID - AI Branding Toolkit (katanaid.com)

12/2025 - Present

Founding Engineer

Remote

- Co-founding and leading 5 engineers to architect a scalable end-to-end branding intelligence backend in Go, leveraging Goroutines to orchestrate 19+ external API calls for real-time asset aggregation and PDF generation.
- Sustained 2.3K+ requests/day under production load, validated via k6 stress testing, by integrating Google Gemini AI for brand asset generation alongside a trust score engine with browser fingerprinting.
- Eliminated schema-related runtime errors by leveraging Ent ORM for a strictly type-safe PostgreSQL layer, enabling reliable large-scale brand data synchronization across distributed API sources.

Vietcombank

05/2025 - 07/2025

Technical Assistant Intern

Hue, Vietnam

- Maintained 99% uptime across 30+ networked branch systems over 3 months by diagnosing hardware, network, and software faults in coordination with IT teams to ensure uninterrupted banking operations.
- Reduced monthly in-branch transactions by 15% by onboarding 100+ customers biweekly onto digital banking services, including mobile app setup, online payments, and account management workflows.

PROJECTS

Draft-Thinker (Live) | *Go, OpenAI API, Qdrant, Redis, Prometheus, Grafana, Docker*

- Reduced LLM inference costs by 91.6% by routing 94% of requests through a drafter model via real-time Shannon entropy analysis of token logprobabilities, achieving 98.2% accuracy on a 518-prompt benchmark.
- Implemented a semantic cache layer with vector search and TTL-based eviction serving repeated queries under 50ms at cosine similarity above 0.95, eliminating redundant inference on cached draft-accepted responses.

Ferrox (Live) | *Rust, Systems Programming, Atomics, Mmap, Criterion*

- Achieved 500ns P99 tick-to-trade latency (verified via HdrHistogram) processing 4.7M simulated orders/second by building a matching engine with zero hot-path allocations via a 1M-slot arena and doubly linked lists.
- Increased throughput by 8.8x by engineering a lock-free SPSC ring buffer with Acquire/Release memory ordering; implemented cache-line padding (64B) and mmap write-ahead log for crash recovery under 1.4ms.

Inyeon (Live) | *FastAPI, LangGraph, ChromaDB, scikit-learn, NumPy, Typer*

- Achieved 100ms median response time by architecting a scalable agentic AI git assistant with 7 agents in FastAPI and LangGraph, orchestrating full pipelines with cost-optimized caching and short-circuiting.
- Attained 100% unit & integration test coverage across 245+ cases, cutting Docker build times by 95% (49s to 2.1s) via RAG-powered ChromaDB analysis across 4 clustering strategies with scikit-learn.

TECHNICAL SKILLS

Languages: Go, JavaScript/TypeScript, SQL, Python, Rust, C/C++, C#, Java, HTML/CSS, MATLAB

Technologies: Nuxt.js, Vue.js, Next.js, React, Node.js, Bun, Express.js, Spring Boot, Chi, Gin, AWS (Amazon Web Services), Docker, PostgreSQL, MongoDB, MySQL, REST API, FastAPI, Django, Flask, LangChain/LangGraph, ChromaDB, scikit-learn, NumPy, Socket.IO, Git, GitHub, Grafana, Prometheus, Redis, CI/CD, Agile/Scrum, Linux